IMAT 모의고사 8회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Artificial-intelligence systems trained on thousands of retinal photographs can now detect diabetic eye disease with accuracy comparable to specialists. Advocates see this as a way to screen millions of patients in regions with few ophthalmologists. However, an algorithm that performs well on images from one hospital's cameras may perform worse on images taken with different equipment or from populations under-represented in the training data. Independent testing in real clinics, rather than reliance on developers' published figures, is therefore essential before such tools are widely adopted.

1. What is the main concern raised in the passage?

A. Retinal photographs are expensive.
B. Specialists oppose new technology.
C. AI systems are slower than ophthalmologists.
D. Diabetic eye disease is rare.
E. AI performance may not generalise beyond the conditions in which it was trained.

2. Which of the following would best address the concern described?

A. Evaluating the algorithm in diverse clinics with different equipment and patient populations.
B. Training the algorithm on images from a single hospital.
C. Publishing more of the developers' own accuracy figures.
D. Replacing all ophthalmologists.
E. Reducing the number of images used for training.

3. Which Nobel Prize-winning discovery of 1901 made it possible for physicians to see inside the body without surgery?

A. X-rays (Röntgen)
B. Insulin
C. The electron
D. Radioactivity
E. Ultrasound

4. Which of these is the official currency used in Italy?

A. Mark
B. Euro
C. Peseta
D. Lira
E. Franc

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "If the experiment is valid, then the results are reproducible. If the results are reproducible, then the paper will be accepted." Which follows?

A. The experiment is valid.
B. If the experiment is valid, the paper will be accepted.
C. If the paper is accepted, the experiment is valid.
D. If the results are not reproducible, the paper is accepted.
E. If the paper is rejected, the experiment may still be valid.

6. An investment of €2000 earns 5% simple interest per year. What is its total value after 3 years?

A. €3000
B. €2100
C. €2500
D. €2315.25
E. €2300

7. Three boxes are labelled "Apples", "Oranges" and "Mixed", but every label is wrong. You may take one fruit from one box. From which box should you take it to identify all boxes correctly?

A. "Apples"
B. "Oranges"
C. "Mixed"
D. It is impossible
E. Any box

8. What number comes next? 2, 3, 5, 7, 11, 13, __

A. 15
B. 17
C. 21
D. 19
E. 16

9. A map uses a scale of 1 : 50 000. Two towns are 6 cm apart on the map. What is the real distance?

A. 8.3 km
B. 30 km
C. 300 km
D. 3 km
E. 0.3 km

Section 3 — Biology (Q10–32)

10. Which of the following structures is found in both prokaryotic and eukaryotic cells?
A. Mitochondria
B. Ribosomes
C. Endoplasmic reticulum
D. Golgi apparatus
E. Nucleus

11. Semi-conservative replication means that:
A. both strands are newly synthesised.
B. both strands of the parent molecule remain together.
C. DNA is copied into RNA.
D. replication occurs only on one strand.
E. each new DNA molecule contains one original strand and one new strand.

12. Which phase of the cell cycle is the longest in most actively dividing human cells?
A. Metaphase
B. Interphase
C. Telophase
D. Prophase
E. Anaphase

13. Which of the following is the correct summary equation for aerobic respiration?
A. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
B. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy
C. $C_6H_{12}O_6 \rightarrow$ 2 lactate
D. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
E. $C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 12H_2$

14. Two heterozygous parents for cystic fibrosis (autosomal recessive) have two children. What is the probability that both children are affected?
A. 9/16
B. 1/4
C. 1/8
D. 1/2
E. 1/16

15. Which of the following best describes the role of decomposers in an ecosystem?
A. Producing oxygen
B. Predation of herbivores
C. Consuming primary producers
D. Fixing atmospheric carbon by photosynthesis
E. Breaking down dead organic matter and recycling nutrients

16. The cells lining the trachea that sweep mucus upwards have:
A. keratin.
B. cilia.
C. flagella.
D. desmosomes only.
E. microvilli.

17. Which of the following elements is present in all proteins but not in carbohydrates?
A. Carbon
B. Phosphorus
C. Hydrogen
D. Nitrogen
E. Oxygen

18. Which part of a neuron receives signals from other neurons?
A. Axon terminal
B. Dendrites
C. Myelin sheath
D. Axon hillock only
E. Node of Ranvier

19. In the ECG, the QRS complex corresponds to:
A. ventricular depolarisation.
B. atrial repolarisation only.
C. closing of the semilunar valves.
D. atrial depolarisation.
E. ventricular repolarisation.

20. A plant heterozygous for three independently assorting genes (AaBbCc) is self-fertilised. What proportion of offspring is homozygous recessive for all three genes?
A. 1/16
B. 1/64
C. 27/64
D. 1/27
E. 1/8

21. Which hormone triggers ovulation?
A. Luteinising hormone (LH)
B. Testosterone
C. Progesterone
D. Oestrogen
E. Prolactin

22. Which of the following describes Chargaff's rule?
A. A = T and G = C
B. A = G and C = T
C. Purines = 25% always
D. A + T = G + C always
E. A = U in DNA

23. Which of the following is an example of passive immunity?
A. Clonal expansion of B cells
B. Maternal antibodies transferred through breast milk
C. Production of memory cells
D. Immunity after vaccination
E. Immunity after recovering from measles

24. Which of the following would be expected in the urine of a person with uncontrolled diabetes mellitus?

A. Proteins
B. Red blood cells
C. Glucose
D. Bilirubin only
E. Platelets

25. What is the number of chromosomes in a human secondary oocyte?

A. 23
B. 22
C. 92
D. 46
E. 69

26. The induced-fit model states that:

A. substrates change enzymes permanently.
B. enzymes are not specific.
C. the active site is rigid and fits the substrate exactly.
D. enzymes have no active site.
E. the active site changes shape slightly when the substrate binds.

27. A point mutation changing GAA (Glu) to GAG (Glu) is:

A. a chromosomal mutation.
B. a frameshift mutation.
C. a nonsense mutation.
D. a silent mutation.
E. a missense mutation.

28. Which of these is NOT a component of the plasma membrane?

A. Peptidoglycan
B. Glycolipids
C. Cholesterol
D. Proteins
E. Phospholipids

29. Bacteriophages are:

A. bacteria that infect viruses.
B. prions.
C. viruses that infect bacteria.
D. protists.
E. fungi.

30. Which organ produces most digestive enzymes that act in the small intestine?

A. Liver
B. Spleen
C. Gallbladder
D. Pancreas
E. Stomach

31. In eukaryotes, where does transcription occur?
A. Golgi apparatus
B. Cytoplasm
C. Nucleus
D. Lysosome
E. Ribosome

32. Which of the following conditions is NOT required for Hardy–Weinberg equilibrium?
A. No mutation
B. No gene flow
C. Random mating
D. Strong natural selection
E. Large population size

Section 4 — Chemistry (Q33–47)

33. What volume does 0.5 mol of an ideal gas occupy at STP (22.4 L/mol)?
A. 44.8 L
B. 2.24 L
C. 11.2 L
D. 22.4 L
E. 5.6 L

34. What is the oxidation number of oxygen in H_2O_2?
A. +2
B. −2
C. +1
D. 0
E. −1

35. Which of the following solutions has the lowest pH?
A. 0.1 mol/L HCl
B. 0.1 mol/L NH_3
C. 0.1 mol/L CH_3COOH
D. 0.1 mol/L NaOH
E. 0.1 mol/L NaCl

36. Which element is a halogen in period 3?
A. Na
B. S
C. Cl
D. Br
E. Ar

37. What is the shape of the methane molecule?
A. Trigonal pyramidal
B. Octahedral
C. Tetrahedral
D. Linear
E. Square planar

38. In the reaction $2KClO_3 \rightarrow 2KCl + 3O_2$, how many moles of O_2 are formed from 4 mol of $KClO_3$?

A. 6
B. 12
C. 3
D. 4
E. 2

39. What is the IUPAC name of $CH_3COOCH_2CH_3$?

A. Butanoic acid
B. Methyl propanoate
C. Ethyl methanoate
D. Ethyl ethanoate
E. Propyl methanoate

40. A 0.2 mol/L solution of $CaCl_2$ contains what concentration of Cl^- ions?

A. 0.1 mol/L
B. 0.8 mol/L
C. 0.2 mol/L
D. 0.6 mol/L
E. 0.4 mol/L

41. Which of the following is a polar molecule?

A. CCl_4
B. CO_2
C. N_2
D. HCl
E. CH_4

42. Which of the following is a Lewis acid?

A. H_2O
B. Cl^-
C. BF_3
D. NH_3
E. OH^-

43. For an exothermic reaction, increasing temperature will:

A. stop the reaction.
B. increase the equilibrium constant.
C. not change the equilibrium constant.
D. decrease the equilibrium constant.
E. make it endothermic.

44. Which pair of compounds are functional group isomers?

A. Ethanol and methanol
B. Propan-1-ol and propan-2-ol
C. Propanal and propanone
D. Butane and methylpropane
E. cis- and trans-but-2-ene

45. How many neutrons are in $^{23}Na^{+}$?
A. 12
B. 23
C. 10
D. 11
E. 22

46. Rusting of iron is an example of:
A. reduction of iron.
B. sublimation.
C. oxidation of iron.
D. neutralisation.
E. precipitation.

47. Which of the following is a saturated hydrocarbon?
A. Propane
B. Ethyne
C. Benzene
D. Propene
E. Ethene

Section 5 — Physics & Mathematics (Q48–60)

48. A stone is thrown upward at 15 m/s. How long does it take to return to the thrower's hand? (g = 10 m/s²)
A. 2 s
B. 1.5 s
C. 4.5 s
D. 3 s
E. 6 s

49. A motor lifts a 50 kg load at constant speed of 2 m/s. What is the motor's power output? (g = 10 m/s²)
A. 500 W
B. 250 W
C. 2000 W
D. 1000 W
E. 100 W

50. In a circuit, a 6 V battery drives current through a 2 Ω and a 4 Ω resistor in series. What is the voltage across the 4 Ω resistor?
A. 3 V
B. 6 V
C. 2 V
D. 1 V
E. 4 V

51. Which type of electromagnetic radiation has the shortest wavelength?

A. Microwaves
B. Radio waves
C. Ultraviolet
D. Visible light
E. Gamma rays

52. A diver goes from the surface to 20 m depth in water. By how much does the pressure increase? ($\rho = 1000$ kg/m³, $g = 10$ m/s²)

A. 2×10^5 Pa
B. 2×10^6 Pa
C. 2×10^3 Pa
D. 10^5 Pa
E. 2×10^4 Pa

53. A net force of 10 N acts on a 2 kg object for 3 s, starting from rest. What is the object's final speed?

A. 5 m/s
B. 10 m/s
C. 15 m/s
D. 6.7 m/s
E. 30 m/s

54. How much energy is required to melt 0.5 kg of ice at 0 °C? (Latent heat of fusion = 3.3×10^5 J/kg)

A. 1.65×10^4 J
B. 3.3×10^5 J
C. 6.6×10^5 J
D. 6.6×10^4 J
E. 1.65×10^5 J

55. What is the value of x if $\ln(x) = 0$?

A. 10
B. 1
C. e
D. -1
E. 0

56. Convert $3\pi/4$ radians to degrees.

A. 270°
B. 90°
C. 45°
D. 135°
E. 120°

57. A test for a disease is positive in 90% of patients with the disease. In a sample of 200 patients with the disease, how many would be expected to test negative?

A. 90
B. 18
C. 180
D. 20
E. 10

58. What is the volume of a sphere of radius 3 cm?

A. 12π cm^3
B. 36π cm^3
C. 27π cm^3
D. 9π cm^3
E. 108π cm^3

59. Factorise: $x^2 + x - 12$.

A. $(x + 3)(x - 4)$
B. $(x - 6)(x + 2)$
C. $(x - 3)(x + 4)$
D. $(x + 6)(x - 2)$
E. $(x + 12)(x - 1)$

60. An alpha particle consists of:

A. two electrons.
B. one proton.
C. one neutron.
D. one electron.
E. two protons and two neutrons.

정답 및 해설

번호	정답	해설
1	E	학습 환경 밖에서의 일반화 문제.
2	A	다양한 임상 환경에서 독립 검증.
3	A	뢴트겐 X선(1901 첫 노벨 물리학상).
4	B	유로.
5	B	삼단 연쇄(가언 삼단논법).
6	E	2000 × 0.05 × 3 = 300 → 2300.
7	C	"Mixed" 라벨 상자는 반드시 단일 과일 → 하나 꺼내면 그 상자 확정, 나머지도 연쇄적으로 결정.
8	B	소수열 → 17.
9	D	6 × 50 000 = 300 000 cm = 3 km.
10	B	리보솜.
11	E	반보존적 복제.
12	B	간기.
13	B	호흡 반응식.
14	E	$(1/4)^2 = 1/16$.
15	E	사체 분해, 영양소 순환.
16	B	섬모.
17	D	질소.
18	B	가지돌기.
19	A	심실 탈분극.
20	B	$(1/4)^3 = 1/64$.
21	A	LH 급증.
22	A	A = T, G = C.
23	B	모유 항체(수동면역).
24	C	포도당뇨.
25	A	제1감수분열 후 n = 23.
26	E	기질 결합 시 활성부위 형태가 약간 변함.
27	D	같은 Glu → 침묵 돌연변이.
28	A	펩티도글리칸은 세균 세포벽.
29	C	세균을 감염하는 바이러스.
30	D	췌장.
31	C	핵.
32	D	자연선택이 없어야 함.
33	C	0.5 × 22.4 = 11.2 L.
34	E	과산화물 −1.
35	A	강산 0.1 mol/L HCl → pH 1.

번호	정답	해설
36	C	염소.
37	C	정사면체.
38	A	2 : 3 → 4 : 6.
39	D	아세트산에틸(에틸 에타노에이트).
40	E	Cl^- 2개 × 0.2 = 0.4.
41	D	HCl.
42	C	BF_3는 전자쌍 수용체.
43	D	발열 반응: T↑ → K↓.
44	C	알데하이드 vs 케톤(C_3H_6O).
45	A	23 − 11 = 12.
46	C	Fe → Fe^{3+} 산화.
47	A	프로페인.
48	D	올라가는 시간 1.5 s × 2 = 3 s.
49	D	P = mgv = 50×10×2 = 1000 W.
50	E	I = 1 A → 4 Ω에 4 V.
51	E	감마선.
52	A	ρgh = 1000×10×20 = 2×10^5 Pa.
53	C	a = 5 → v = 15 m/s.
54	E	0.5 × 3.3×10^5 = 1.65×10^5 J.
55	B	e^0 = 1.
56	D	135°.
57	D	10% × 200 = 20.
58	B	4/3 π × 27 = 36π.
59	C	(x−3)(x+4).
60	E	양성자 2 + 중성자 2(헬륨 핵).